

Factor 10 mostly switches the closure off

GABLS1 factor-10 sensitivity | 21 September 2026 | 12.5 m, WENO9, one interior face, 300 s filter, 9 h, seed 123. All earlier factor-1/2 results are retained separately.

Result: factor 10 largely removes the added surface-layer mixing after spin-up. Resolved turbulence then resembles the earlier no-closure run. In 8-9 h, first-face w^2 is $0.064671 \text{ m}^2/\text{s}^2$, versus 0.00817056 for factor 1, 0.00875396 for factor 2, and 0.0635343 without closure. Factor 10 is 7.92 times factor 1 and only 1.79% above the no-closure value. This is recovery of resolved turbulence, not proof of accurate total transport.

Mechanism: the multiplier acts only inside the signed local deficit, $\max(0, 1 - a F_{\text{resolved}}/F_{\text{target}})$, for momentum and heat. With $a=10$, aligned resolved transport carrying 10% of the target can satisfy the deficit. Countergradient transport increases it. Crediting ten times the resolved flux assumes a missing contribution nine times as large; the experiment does not measure that contribution. Physical covariance diagnostics are never multiplied.

The saved fractions explain the response. In the final hour, momentum viscosity is exactly zero at 99.9934896% of sampled horizontal point-times; heat diffusivity is zero at 100%. The corresponding raw-zero-deficit and guard-valid-zero-deficit fractions agree. Momentum and heat guards remain valid at 100%, with zero coefficient-cap activity, so this is deficit satisfaction rather than guard failure. Each value averages 60 one-minute samples over all 32×32 horizontal points. It does not imply continuous switch-off between samples: true averaged SGS heat flux is tiny but nonzero, $-1.26136\text{e-}08 \text{ K m/s}$. These six fractions were not saved for factors 1/2 and remain unavailable there.

The closure acts during spin-up. During 0-1 h, mean viscosity is $0.5168 \text{ m}^2/\text{s}$ and heat diffusivity is $0.6870 \text{ m}^2/\text{s}$; zero-coefficient fractions are about 81.54% and 82.98%. By 1-2 h, momentum is zero at 99.9837% of sampled points and heat at 100%. Small intermittent reactivations remain later. In 8-9 h the mean viscosity is $2.72361\text{e-}05 \text{ m}^2/\text{s}$, versus 1.32118 and 1.11797 for factors 1 and 2; sampled mean heat diffusivity is zero.

Moments and energy: factor-10 first-face w^3 is $+0.00977604 \text{ m}^3/\text{s}^3$ and skewness $+0.59443$, compared with negative values for factors 1/2 and $+0.56311$ without closure. Final-hour peak w^2 is 0.0877192 , and integrated resolved TKE is $35.1207 \text{ m}^3/\text{s}^2$, 28.83% above factor 1. The response persists in 7-8 h: first-face w^2 is 0.0668137 , skewness $+0.58543$, and integrated TKE 35.2916 . Neighboring hours and one seed are not an uncertainty estimate.

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Transport and exchange: final-hour SGS carries 0.012852% of first-face u-momentum covariance-plus-SGS transport, versus 88.66% and 86.82% for factors 1/2. Factor-10 u^* is 0.279088 m/s and surface sensible heat flux -15.2328 W/m², close to the no-closure values 0.279164 and -15.2131. The plots' quantity called total is covariance plus actual constitutive SGS. It is NOT the scheme-native WENO advective flux plus SGS: the numerical correction to covariance has not been reconstructed. Agreement of that plotted quantity with a reference cannot validate the missing numerical contribution.

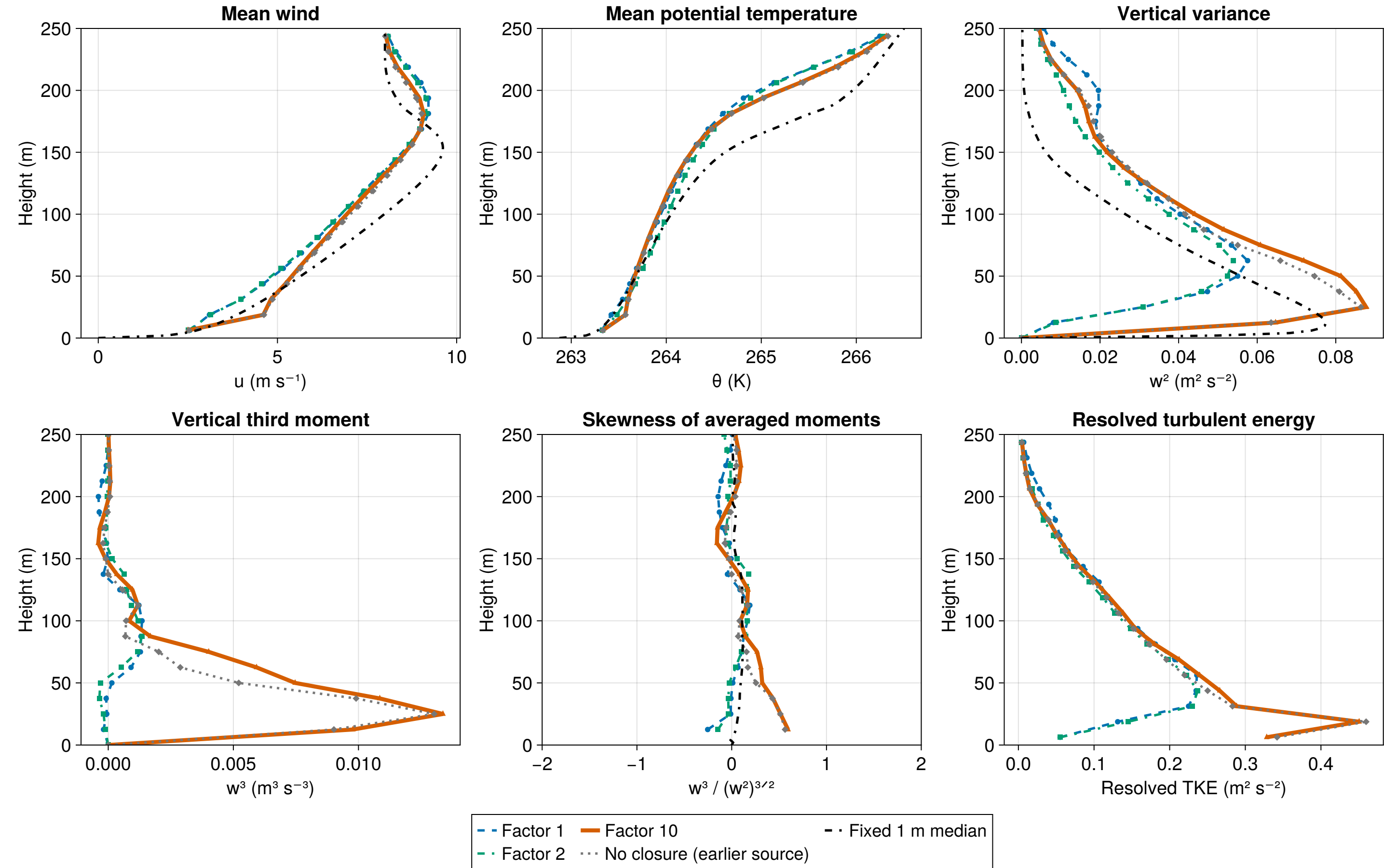
Reference comparison: relative to factor 1, final-hour native-level RMS errors against the fixed 1 m LES median fall 17.33% for u , 9.354% for θ , and 27.43% for w^2 . Nevertheless, factor-10 errors remain larger than the no-closure errors for all three quantities in 8-9 h. The reference is an LES ensemble median, not observations. This strong sensitivity chiefly approaches the no-closure regime; it does not establish factor 10 as a calibrated or faithful correction.

Comparison integrity: factor10 passed separate strict admission (1 admitted, 0 rejected), with 19 native profiles, 541 exact series times, finite values, supported-face SGS and covariance-plus-SGS consistency. Factors 1/2 retain their original admission. Breeze physics commit is identical; factor10's evaluation revision adds read-only diagnostics, with source-specific GPU validation (4,462 checks). Initial native profiles match exactly. Fresh factor1 already reproduces every corrected historical one-face300 profile and series value bit for bit. The earlier no-closure run is labeled as source context. Job7339 used a non-login wrapper and its durable child record verifies exit zero. The preserved older factor2 scheduler discrepancy is documented separately.

Definitions: plotted 8-9 h profiles average the two true half-hour means ending at 30600 and 32400 s; 7-8 h uses 27000 and 28800 s. Series metrics average samples strictly after the hour start through its end. Skewness is the ratio of averaged third moment to averaged variance^(3/2), omitted for $w^2 < 1e-6$. Native model heights are retained; only the fixed 1 m median is interpolated for unweighted RMS errors over $0 < z \leq 200$ m. No WENO flux reconstruction or additional experiment is included.

Factor 10: mean profiles and resolved turbulence

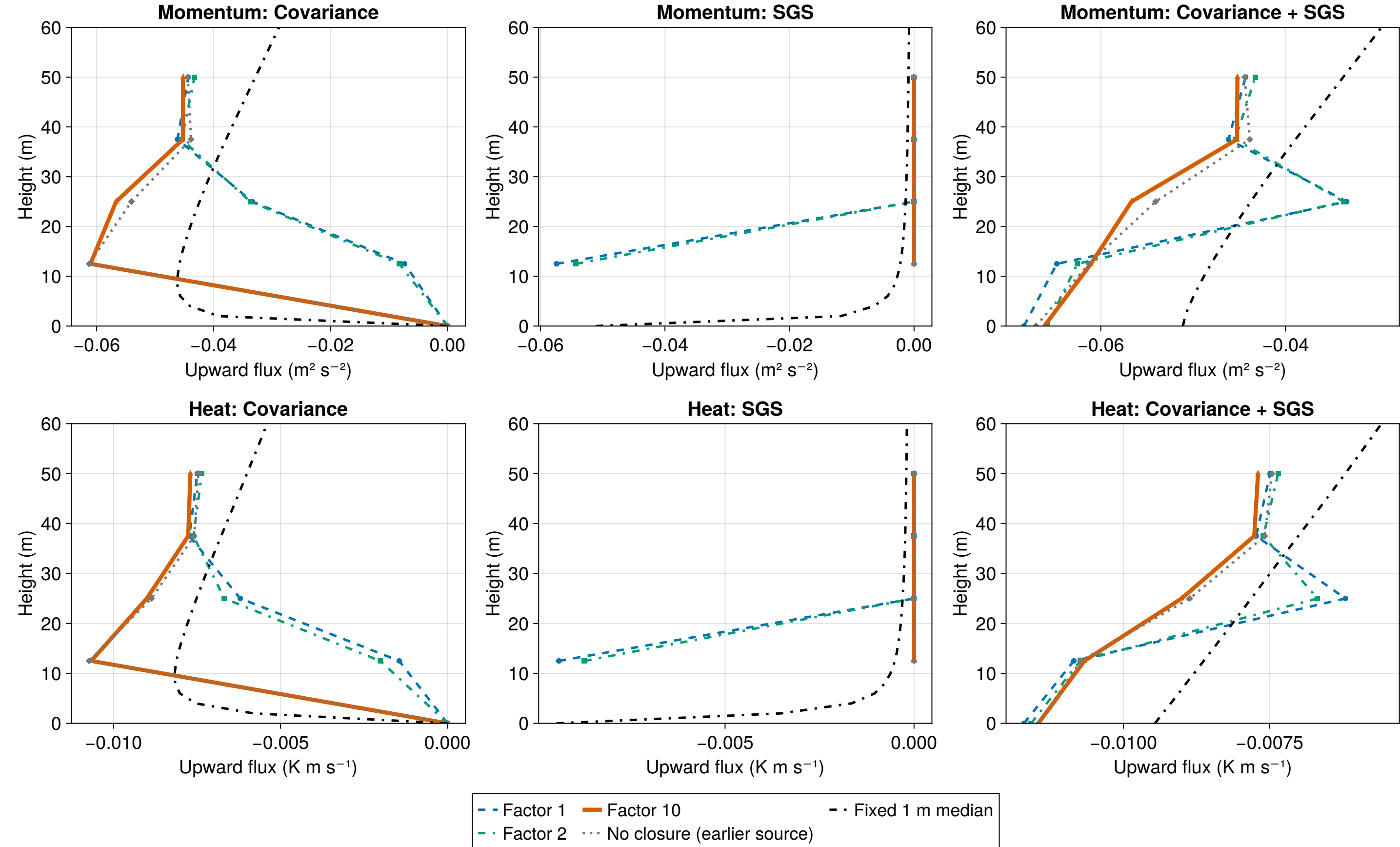
GABLS1 · 12.5 m · WENO9 · one face · 300 s filter · 8–9 h mean



Factor 10 is vermillion, solid, triangles. Reference medians shown where available.
Native heights and physical moments retained; the added numerical-flux credit is an assumption.

Covariance and SGS transport near the surface

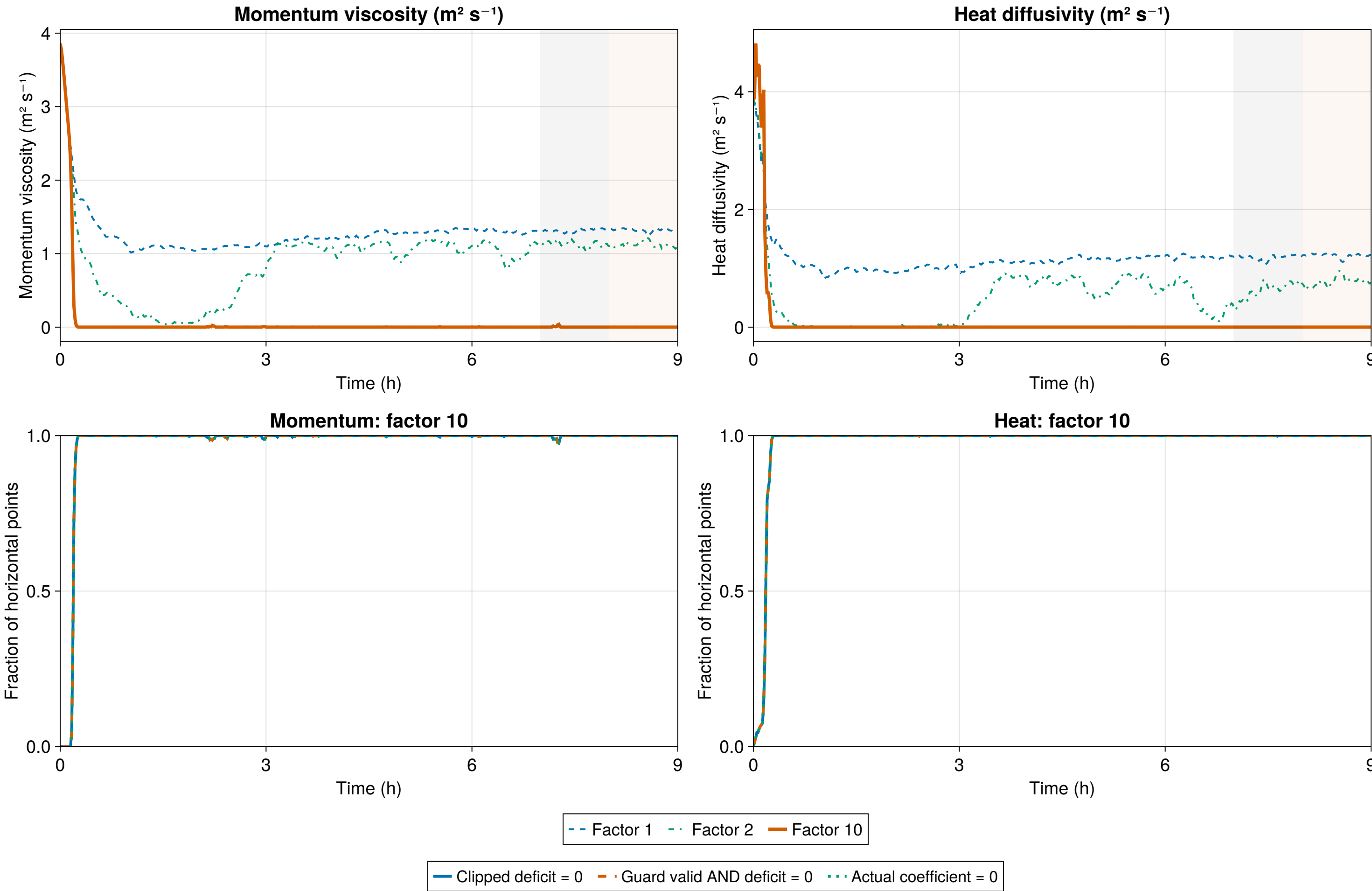
Same final-hour average · native flux faces · physical fluxes remain unscaled



Interior total = covariance + actual SGS. This is NOT scheme-native WENO transport: its numerical correction is missing.
Wall total is prescribed; artificial SGS wall zeros omitted. Fixed 1 m medians shown where available.

What the factor changes: coefficients and local switch-off

One supported face at 12.5 m · coefficients from all three factors · switch-off fractions saved only for factor 10



All fraction denominators include every horizontal point. Exact overlap is meaningful; quantitative values are in the report.
Active/guard validity is separate from positive mixing. These six fractions were not saved for factors 1 and 2.